

Doc. 316 – FFGFT and the Riemann Zeta Function

Reach and Limit: Two Identities, Four Exclusions

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Preliminary: the answer in short

Result

The Riemann hypothesis is an **arithmetic** matter. The torus and ξ contribute nothing to its solution. The result of this document is a *negative* one – three obvious geometric approaches are excluded with reasons given.

The blanket formulation would be too coarse in one respect, hence the refinement.

What remains purely a matter of numbers. Where the zeros lie, how they stand relative to one another, how one grasps them individually – that is carried by arithmetic, by the primes. Three geometric approaches were tested and all are negative: no compact geometry can carry the zeros as its spectrum (Section .3, at theorem level), ξ does not shift the critical line (Section .2), and the ξ ladder discretises nothing (Section .5, out-of-sample test negative).

What is nevertheless not merely a matter of numbers. Two things. First, the Riemann zeta is a *factor* in the spectrum of the $\mathbb{Z}^4$ torus – an identity, not a picture (Section .2). The Riemann hypothesis is thereby a statement about a geometric object of the framework; only one that the geometry cannot answer. Second, the form of the zero density says *where* the question belongs: $N(T)$ grows like $T \ln T$, and no Laplacian on any compact manifold can deliver that – it requires a dilation structure, hence the unrolled domain. The geometry indicates the address without supplying the answer.

The striking thing is precisely that both occur together: the numbers demonstrably sit in the torus spectrum, but the torus cannot generate them.

The yield. The dead ends are closed, with reasons rather than conjecture – and four positive findings stand alongside them: two identities of the framework (the factorisation of the torus spectral zeta and the symmetry position of the Casimir interval) and two measurements on the prime structure (the bicoherence-measured cross-term situation and its physical evidence in the harmonic series).

.1 Starting point and claim

[Q] The corpus mentions the Riemann hypothesis in exactly one place: Doc. 024 notes, in an AI/network context, that the “number space” has dimensions “modulated by the Riemann hypothesis and ζ functions”. That is a conceptual marginal remark without an epistemic marker and without derivation. A worked-out connection does not exist in the corpus.

[K] At the same time Doc. 314 contains material that touches the ζ function directly: Epstein zeta, Casimir point, heat kernel on T^4 . This document examines systematically what follows.

Claim. It proves nothing about the Riemann hypothesis and does not claim to. Three things are clarified: which connections hold as identities **[K]**, which path carries constructively **[B]**, and which obvious paths are *demonstrably* not passable **[X]**. The exclusion is the most valuable part – it saves work on dead ends.

.2 Two findings that hold as identities

The Riemann zeta is a factor of the torus spectral zeta

[K] Doc. 314 carries the closed form of the $\mathbb{Z}^4$ spectral zeta (verified threefold there; confirmed here against the direct sum with analytic tail correction to 10^{-7} through 10^{-12}):

$$Z_{\mathbb{Z}^4}(s) = 8(1 - 4^{1-s})\zeta(s)\zeta(s-1).$$

This is not an analogy but an identity. The zeros of the torus spectral zeta therefore have three sources:

Source	Location of the zeros
$\zeta(s) = 0$	$\Re(s) = 1/2$ (non-trivial – the RH)
$\zeta(s-1) = 0$	$\Re(s) = 3/2$ (shifted by 1)
$1 - 4^{1-s} = 0$	$s = 1 + 2\pi i k / \ln 4$ (lattice factor)

On this reading the Riemann hypothesis is a statement about the spectrum of the $\mathbb{Z}^4$ torus – about a geometric object of the framework, not merely about a number series.

The Casimir point lies symmetric to the critical line

[K] Doc. 314 computes the ζ -regularised vacuum energy $E = \pi\zeta_M(-1/2)$ and finds the order reversal: D4 minimises the Epstein zeta for $s > 0$, at $s = 0$ all lattices are equal, and at $s = -1/2$ the $\mathbb{Z}^4$ torus lies lower (-0.9321 against -0.8686).

The Casimir point $s = -1/2$ has, under the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$, the mirror point $s = 3/2$. The axis of symmetry:

$$\frac{-1/2 + 3/2}{2} = \frac{1}{2} \quad \text{– exactly the critical line.}$$

The physically occupied interval $[-1/2, 3/2]$ has the critical line as its midpoint: consistency between physics and arithmetic.

[X] No proof follows from this – the critical line is fixed by the functional equation anyway. Nor can ξ shift it: a perturbation of order $\xi \sim 10^{-4}$ (even with the RG amplifier 100ξ) does not reach an axis fixed by a functional identity.

.3 The decisive obstruction: Weyl against Riemann-von Mangoldt

The obvious move would be to seek an operator whose eigenvalues are the zeros (Hilbert-Pólya programme), and to deform $T^4/\mathbb{Z}_3$ suitably for that purpose. **This entire branch is excluded – at theorem level, not as intuition.**

[Q] Weyl's law. For every compact Riemannian manifold of any dimension d , asymptotically $N(\lambda) \sim C_d \text{Vol} \lambda^{d/2}$ – a pure power. Curvature, boundary and orbifold points change only lower orders, not the leading term.

[Q] Riemann-von Mangoldt (proved).

$$N(T) = \frac{T}{2\pi} \ln \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + O(1/T).$$

[K] The comparison via the local exponent $d \ln N / d \ln T$:

T	$N(T)$	local exponent $d/2$
10^3	$6.48 \cdot 10^2$	1.2457
10^5	$1.38 \cdot 10^5$	1.1153
10^8	$2.48 \cdot 10^8$	1.0642
10^{12}	$3.95 \cdot 10^{12}$	1.0403
10^{20}	—	1.0233

The exponent is $1 + 1/\ln(T/2\pi)$ – it falls monotonically towards 1 but is *never constant*. Range over 17 decades: 0.22. The best power fit in the window $[10^3, 10^6]$ deviates systematically by up to 11 %.

Exclusion at theorem level

[X] There is no compact Riemannian manifold – of whatever dimension, curvature, topology or orbifold structure – whose Laplace spectrum is the Riemann zeros. Reason: $T \ln T$ is not a power $T^{d/2}$.

Consequence. The branch “deform $T^4/\mathbb{Z}_3$ suitably” is settled completely, not only in the flat case. The extra $\ln T$ is the signature of a **scale degree of freedom**: a dilation direction with logarithmically growing volume. That is not a Laplacian on a manifold.

Not a curvature problem. What generates logarithmic lengths is a dilation structure – the multiplicative group $\mathbb{R}_+^*$ – not curvature; the Berry-Keating operator $H = xp$ is the generator of dilation, not of a hyperbolic flow. The Weyl obstruction therefore applies independently of any metric and renders the curvature question moot.

.4 The constructive link – and why it does not carry

If the zero density requires a scale degree of freedom, the responsible domain is the **unrolled** one (Doc. 315) – multiplicative, logarithmic, with dilation as its natural operation.

The approach

[Q] Berry-Keating (1999) [1]. The semiclassical count of the dilation generator $H = (xp + px)/2$ in a phase space with a minimal cell:

$$N(E) = \frac{1}{2\pi} [E \ln(E/(l_x l_p)) - E + l_x l_p].$$

With $l_x l_p = 2\pi$ this becomes exactly the Riemann-von Mangoldt formula (verified: difference constant $0.125 = 1 - 7/8$, a pure boundary term, over $T = 15 \dots 5000$).

The obvious chain would then be: Doc. 314 Ch. D1 carries $\lambda_4 \cdot m = 2\pi$ as an exact position relation; in the unrolled domain the quantity conjugate to scale is the dilation momentum; hence FFGFT supplies the cell condition.

Why the chain does not carry

[X] Stage 1 – the cell condition is empty of content. $l_x l_p = 2\pi\hbar$ is, in natural units, the Planck cell, and follows from the canonical quantisation of every conjugate pair. That $\lambda_4 m = 2\pi$ has the same form is no independent finding – the de Broglie relation is the statement $\hbar = 1$. Moreover the count is logarithmically insensitive to the cell (a factor 2 wrong $\Rightarrow N$ changes by $\sim 1\%$); a “hit” would not have been a test in any case.

[X] Stage 2 – the actual content fails quantitatively. Berry-Keating requires something sharper than the cell: two *independent* cutoffs $x > l_x$ and $p > l_p$ whose product is the Planck cell. The FFGFT double cutoff makes this testable:

Quantity	Value
UV: $L_0 = \xi \ell_P$	$2.155 \cdot 10^{-39} \text{ m}$
IR: $L^* = 4\tilde{\lambda}_e/\xi^{10}$	$8.698 \cdot 10^{26} \text{ m}$
Ratio L^*/L_0	$4.04 \cdot 10^{65}$
required	$2\pi \approx 6.28$
Shortfall	65 decades

[X] Stage 3 – a finite window against infinitely many zeros. $\ln(L^*/L_0) = 151.06$, that is 16.93 ξ ladder rungs; the Berry-Keating count within the window gives ~ 1816 modes. A finite scale window carries finitely many modes; the zeros are countably infinite. Even if stages 1 and 2 held, at most an initial segment would result – and an initial segment is not a spectral identity.

Marginal observation, explicitly not booked: 16.93 lies close to 17. Both cutoffs carry their own uncertainties – the prefactor 4 is open per R73, ℓ_P depends on G . Without an error budget this is not assessable under P35.

What the chain achieves

[B] What does carry is a structural statement, and it is not little: that the counting function has the form $(T/2\pi) \ln(T/2\pi)$ at all requires a **dilation generator**. A Laplacian on a compact manifold cannot deliver it (Section .3). The substantive statement is the *identification of the structure* – the unrolled domain carries the dilation – not the numerical value 2π .

.5 The missing boundary condition – test negative

$H = xp$ has *continuous* spectrum. The counting function fixes the mean density, not the individual zeros. What is missing is the boundary condition that turns the continuum into a discrete spectrum.

The natural candidate. The ξ ladder generates a discrete subgroup $\Gamma_\xi = \{\xi^n\} < \mathbb{R}_+^*$; the quotient $\mathbb{R}_+^*/\Gamma_\xi$ is a *circle* of circumference $L = \ln(1/\xi) = 8.9227$. Admissible dilation frequencies would then be multiples of $2\pi/L = 0.7042$.

Test discipline

[K] Proximity arguments require three controls (P35): a null distribution over random periods in the *same* range; artefact control, because periods larger than the data range automatically produce small scores; and an out-of-sample check with look-elsewhere correction.

The artefact control is mandatory: periods above the data range automatically produce low scores, because all quotients round to the same integer. At period $2\pi/|\ln(74/75)| = 468$, for

instance, all zeros lie below 102; a score near zero means nothing there. The null distribution must cover the same range of periods as the candidates, or spurious signals arise.

Result

[K] Candidates within the admissible range (null distribution: mean 0.2502):

Candidate	Period	Score	p value
$2\pi / \ln(1/\xi)$	0.7042	0.2812	0.92
$2\pi / \ln 75$	1.4553	0.2043	0.005
$\ln(1/\xi)$	8.9227	0.2891	0.94
2π	6.2832	0.2366	0.18

The borderline candidate $2\pi / \ln 75$ was checked out of sample (zeros 31–50): score 0.2686, $p = 0.81$; look-elsewhere corrected $p = 1.0$.

Boundary condition

[X] The ξ ladder does not supply the missing boundary condition. No ξ -derived period candidate survives the out-of-sample check.

Why this was structurally to be expected

[K] A single-period compactification produces an *equidistant* lattice. The zeros, however, show GUE statistics with level repulsion (Montgomery-Odlyzko [11, 10]): mean normalised spacing 0.9815 (should be ~ 1), fraction of spacings < 0.3 equal to 0.000 – against ~ 0.04 for GUE and ~ 0.26 for a lattice-like spectrum. A circle cannot deliver GUE in principle.

The missing boundary condition is carried by arithmetic (the primes), not by geometry.

.6 The harmonic approach: ξ as a Tonnetz point

The preceding sections treat ξ as a *magnitude* – a parameter whose smallness decides its effect. A second approach asks instead about its *harmonic structure*, that is, about the position of ξ in the prime lattice. It leads to a different point of contact with the theory and yields a sharper exclusion.

The starting observation

[K] The prime factorisation

$$\frac{1}{\xi} = 7500 = 2^2 \cdot 3 \cdot 5^4$$

places ξ as a point in Euler's Tonnetz with coordinates $(a_2, a_3, a_5) = (-2, -1, -4)$. This is the same lattice structure as in the control case of the musical spiral: the primes 2, 3, 5 as axes of the 5-limit space. Equivalently in logarithmic form:

$$\ln(1/\xi) = 2\ln 2 + \ln 3 + 4\ln 5 = 8.922658\dots$$

The harmonic reading suggests a different point of contact than the Laplace operator. A Tonnetz carries three incommensurable logarithmic frequencies $\ln 2, \ln 3, \ln 5$; three independent frequencies generate a quasi-periodic, precisely not equidistant spectrum. The approach therefore touches the arithmetic of the ζ function directly.

The responsible point of contact: Weil's formula

[Q] Weil's explicit formula couples zeros and primes:

$$\sum_{\rho} h(\gamma) = (\text{archimedean term}) - 2 \sum_{p,k} \frac{\ln p}{p^{k/2}} \hat{h}(k \ln p).$$

The dual *length spectrum* of the zeros is therefore

$$\mathcal{L} = \{k \ln p : p \text{ prime}, k = 1, 2, 3, \dots\},$$

that is, multiples of logarithms of **single** primes. Whether ξ appears there is directly testable: the Fourier transform of the zero density must show peaks at the lengths in $\mathcal{L}$.

Measurement

[K] Tested is $F(u) = |\sum_n e^{i\gamma_n u}|/N$ for the first 100 zeros; the background is 3000 random u in the same range (median 0.0304, 95 % threshold 0.2005, 99 % threshold 0.2572).

u	$F(u)$	Assignment
1.09861	0.2321	$\ln 3$ (peak, 95 %)
1.60944	0.2651	$\ln 5$ (peak, 99 %)
1.94591	0.2625	$\ln 7$ (peak, 99 %)
2.39790	0.2610	$\ln 11$ (peak, 99 %)
2.56495	0.2676	$\ln 13$ (peak, 99 %)

The primes are visible; the method works. The same measurement for ξ -derived lengths:

u	$F(u)$	Quantity
8.92266	0.1775	$\ln(1/\xi) = 2 \ln 2 + \ln 3 + 4 \ln 5$
4.46133	0.0777	$\ln(1/\xi)/2$
4.31749	0.0384	$\ln 75 = \ln 3 + 2 \ln 5$
2.23066	0.0189	$\ln(7500)/4$
0.70418	0.1313	$2\pi / \ln(1/\xi)$

None reaches the 95 % threshold.

Resolution limit. The measurement becomes assessable only once u spreads the phases over the full circle, that is $u \gg 2\pi/(\gamma_{\max} - \gamma_{\min}) = 0.02824$. For $u \rightarrow 0$ all phases coincide and $F(u) \rightarrow 1$, independently of any structure. Candidates below this limit – for instance $\ln(75/74) = 0.01342$ – are *not measurable* with 100 zeros; they yield neither confirmation nor refutation.

The structural reason

Combination against factorisation

Tonnetz and Euler product share the prime axes but use them in **opposite** ways. The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ logarithmises to a sum over prime powers: each prime stands on its own, without cross terms – hence $\mathcal{L}$ contains only $k \ln p$. The Tonnetz lives on the opposite: an interval is $2^{a_3} 3^{b_5} 5^c$ with several non-zero exponents, and it is precisely the mixing that generates the commas. ξ is such a mixed point.

[X] $\ln(1/\xi)$ is therefore a *sum* of lengths and not a length; sums do not appear as peaks in the trace formula. The nearest genuine length is $3 \ln 19 = 8.83332$ at distance 0.08934 – without significance, since proximity within the dense length spectrum says nothing.

Sharpness of the exclusion. It is structural, not a matter of magnitude: it holds unchanged if ξ were of order 1. What matters is not the value of ξ but that $1/\xi$ is composite. This section therefore carries a stronger reason than any estimate based on the smallness of ξ .

Remaining test surface. The *individual* axes $\ln 2, \ln 3, \ln 5$ are lengths in the Weil spectrum – but they are so for any theory containing primes at all; the factorisation of 7500 selects nothing. A statement would arise only if the *exponents* $(2, 1, 4)$ produced a weighting within the spectrum. The following subsection tests this.

Does the 5-limit include the higher ones?

The limit extension of Test B raises the converse question: can it be shown that the 5-limit already contains the higher limits? The answer comes in five gradations, and the gradation repeats the pattern of the whole document (`z9_limit_einschluss.py`).

[K] (1) Exactly: no, provably. $\ln 7 = a \ln 2 + b \ln 3 + c \ln 5$ with rational a, b, c would mean, after clearing denominators, $7^q = 2^A 3^B 5^C$ – contradicting unique factorisation. This holds for every prime above the limit, hence for infinitely many.

[K] (2) As a dense approximation: yes, arbitrarily well. Since $\ln 2, \ln 3, \ln 5$ are linearly independent over $\mathbb{Q}$, the lattice lies dense in $\mathbb{R}$:

N	Error	Cents	Exponents	Tenney
2	$2.82 \cdot 10^{-2}$	48.77	$(2, 2, -1)$	$5.6 \cdot 10^{-3}$
4	$7.97 \cdot 10^{-3}$	13.80	$(-1, -2, 3)$	$4.4 \cdot 10^{-4}$
6	$4.45 \cdot 10^{-3}$	7.71	$(-5, 2, 2)$	$1.4 \cdot 10^{-4}$
8	$2.29 \cdot 10^{-4}$	0.40	$(1, 7, -4)$	$3.7 \cdot 10^{-7}$

The error of the best approximation $(1, 7, -4)$ is exactly $\ln(4375/4374)$ – **the ragisma**. The coarse approximation $(6, -2, 0) = \ln(64/9)$ has error $\ln(64/63)$, **Archytas' comma**. Both are known from the Euler control case: there as near-closures in the 7-limit, here as the error of the 5-limit approximation to the 7-axis. It is the same quantity read from the opposite direction.

[K] (3) At finite resolution: yes, indistinguishable. The peak width of the measurement in Section .6 is $2\pi/(\gamma_{\max} - \gamma_{\min}) = 0.0021$; the ragisma, at $2.29 \cdot 10^{-4}$, is only 0.11 peak widths:

Location	u	Power
$\ln 7$ exact	1.945910	2.144
approximation $(1, 7, -4)$	1.945682	2.144
coarse $(6, -2, 0) = \ln(64/9)$	1.961659	1.417
control $\ln 7 + 0.05$	1.995910	0.042

The fine approximation is indistinguishable from $\ln 7$, the coarse one falls off, and the control point lies in the noise. At this resolution the 5-limit does effectively contain the 7-axis – a statement about the measuring procedure, not about the arithmetic.

[K] (4) Amplitude-weighted: no, and clearly so. The Tenney weight of the approximation $(1, 7, -4)$ is $3.7 \cdot 10^{-7}$ against $1/7$ for $\ln 7$ as an axis of its own – a factor $3.9 \cdot 10^5$. This also explains *why* the approximation can be so accurate: it requires large exponents, and large exponents mean small weight. Accuracy and weight run opposite to one another.

[B] (5) By tempering: yes, but by stipulation. Declaring the ragisma equal to 1 makes $7 = 2 \cdot 3^7/5^4$ exact in the tempered system; likewise, tempering out Archytas' comma makes the approximation $64/9$ exact.

Inclusion does not arise from structure

The 5-limit does not include the higher limits – unless approximation, limited resolution or tempering is allowed to count as inclusion. This is the same pattern as everywhere in this document: inclusion arises through *rationalisation* (tempering) or through *blur* (resolution), never through structure. Likewise Euler's spiral closes only by tempering, and likewise coupling in the Weil spectrum sits only at exact prime powers (Section .6).

For ξ nothing new follows, but a confirmation: that ξ is a pure 5-limit point cannot be cured by any extension – and the approximations that might replace an extension carry no weight.

Measurement instead of assertion: the bicoherence test

The statement that the Euler product has no cross terms was up to this point argued from the *form* of the product. It can be measured – with the procedure responsible for it and with built-in controls.

Setup. The sequence of zeros is taken as a point measure; its transform in the length variable is $F(u) = \sum_n e^{i\gamma_n u}$. Averaged over height blocks, bispectrum and bicoherence [6] follow as in Section .10. The zeros are computed here via the Riemann-Siegel formula: 2469 of them up to $t = 3000$ (Riemann-von Mangoldt expects 2468.6), residual error $7.5 \cdot 10^{-3}$ in γ , corresponding to 0.02 rad of phase error at $u \approx 2.6$.

Test plan. The Weil spectrum contains $k \ln p$. This yields predictions with positive and negative controls: if the sum of two lengths is again a length – that is, a prime power – coupling must appear; if it leads to a composite number, none may appear.

Pre-check. The power of $F(u)$ shows the individual lengths $\ln 2 \dots \ln 13$ all above the 99 % threshold. The background must be cleaned of the 13 genuine lengths in the window – random u hit one of them with about 2 % probability and otherwise form the outlier tail (median 0.055 against a 99 % threshold of 0.881 after cleaning).

Result (24 height blocks of 102 zeros each; critical value from 2000 phase-randomised realisations per pair):

Pair	Sum	b^2	$b_c(99\%)$	Verdict	expected
($\ln 2, \ln 2$)	$\ln 4 = 2^2$	0.8088	0.2612	coupling	coupling
($\ln 3, \ln 3$)	$\ln 9 = 3^2$	0.8462	0.2214	coupling	coupling
($\ln 2, 2 \ln 2$)	$\ln 8 = 2^3$	0.7586	0.2432	coupling	coupling
($\ln 2, \ln 3$)	$\ln 6$ comp.	0.0253	0.1832	none	none
($\ln 2, \ln 5$)	$\ln 10$ comp.	0.0252	0.1912	none	none
($\ln 3, \ln 5$)	$\ln 15$ comp.	0.0175	0.1718	none	none

Cross terms: measured

Six of six predictions met, with a contrast of a factor 30 to 50 between prime powers and composite numbers. Coupling occurs exactly where the sum of two lengths is again a length, and nowhere else. **[K]** The cross-term statement is thereby measured, not merely argued structurally.

Consequence for ξ . $\ln(1/\xi) = 2 \ln 2 + \ln 3 + 4 \ln 5$ is a sum over *three different* primes and leads to the composite number 7500. By the pattern measured here, no coupling can sit there – which the independent Fourier finding of Section .6 ($F = 0.1775$, below every threshold) confirms. Open item Z-4 is thereby closed, and the exclusion of Section .6 now rests on a measurement rather than on an argument from form.

Can a weighting exist? Three tests

The remaining test surface read: a statement would arise only if the exponents (2, 1, 4) produced a weighting in the Weil spectrum. Three independent tests decide this question negatively (z7_gewichtung_unmoeglich.py).

[K] Test A – weighting strength. Higher harmonics have smaller amplitude; the standard measure is Tenney height, weight $\sim 1/(nd)$. An exponent makes the strength free:

$$w = (2^{|a|} 3^{|b|} 5^{|c|})^{-\alpha},$$

with $\alpha = 0$ unweighted and $\alpha = 1$ Tenney. The weight sum has the closed form $\prod_p [1 + 2p^{-\alpha}/(1 - p^{-\alpha})]$ and converges for every $\alpha > 0$:

α	Weight sum	eff. point count	Character
0.10	4,284	23,843	blindly dense
0.30	245.8	1,767	middle
0.50	56.9	360	middle
1.00	9.0	36	axes only

Both extremes are unsuitable: unweighted, the lattice hits everything and selects nothing; at Tenney weighting only the axes remain. What matters is the middle around $\alpha \approx 0.3$ with several hundred to a thousand carrying points. The selection test across the whole range:

α	$M(\text{Weil})$	$M(\text{random})$	Ratio	p
0.15	2.1205	2.0943	1.013	0.27
0.30	0.5344	0.5210	1.026	0.31
0.50	0.1414	0.1493	0.947	0.64
1.00	0.0114	0.0226	0.503	0.93

No α produces selection; the ratio stays at or below one. The weighting strength is not the problem.

[K] Test B – limit extension. The 5-limit is the simple form of the Tonnetz; extended Tonnetze take in further prime axes, 7 being the next. Tested at $\alpha = 0.3$; primes up to the respective limit are axes and hence excluded as trivial.

Limit	Axes	Points	Ratio	p
5	3	2,197	1.049	0.28
7	4	28,561	1.015	0.38
11	5	371,293	1.048	0.11

Weightier than the p values is the structural point: every extension merely shifts the boundary, it closes no gap.

Limit	trivially covered	entirely absent
5	2, 3, 5	all $p \geq 7$
7	2, 3, 5, 7	all $p \geq 11$
13	2 ... 13	all $p \geq 17$

A Tonnetz with limit L covers the finitely many primes $p \leq L$ by definition and the infinitely many $p > L$ not at all – there is nothing in between. Weil’s formula, however, sums over all primes. The limiting case settles it: a Tonnetz with infinite limit would no longer be a structure; it would contain every prime as an axis, that is, the Euler product itself – without additional content and without any relation to ξ .

In addition, $\xi = 2^2 / (2^4 \cdot 3 \cdot 5^4)$ has exponent zero on all axes from 7 onwards. An extension enlarges the lattice but does not change the position of ξ within it; ξ remains a pure 5-limit point.

[K] Test C – rigidity. This is the actual reason, and it holds independently of any numerical computation. The weights of the explicit formula are not free parameters but a consequence of the Euler product:

$$\log \zeta(s) = \sum_{p,k} \frac{1}{k} p^{-ks},$$

from which $\ln p/p^{k/2}$ follows by differentiation and shifting to $s = 1/2$. There is no place at which an additional factor could be inserted without altering the Euler product itself. Setting $\prod_p (1 - g p^{-s})^{-1}$ with $g \neq 1$ by way of trial produces a different Dirichlet series with a shifted abscissa of convergence and *without* a functional equation.

Why no weighting is possible

The functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ is precisely what defines the critical line. A weighting meant to explain the location of the zeros thereby destroys the object whose zeros it seeks to explain. **[X]**

Test C carries on its own; A and B show in addition that the numerical routes yield nothing either – neither through the choice of weighting nor through extended Tonnetze. Open item Z-1 is therefore not unanswered but **decided in the negative**.

.7 Balance

Statement	Status
ζ is a factor of the torus spectral zeta	[K] verified
Casimir interval has critical line as axis	[K]
Zero density requires dilation generator	[B] structure
Cell condition as an independent finding	[X] only $\hbar = 1$
Double cutoff as BK truncation	[X] 65 decades
Compact manifold with zero spectrum	[X] Weyl
ξ ladder discretises the zeros	[X] test negative
$\ln(1/\xi)$ in the Weil length spectrum	[X] a sum, not a length
Cross terms in the Euler product	[K] measured by bicoherence
5-limit contains higher limits	[X] only approximated/tempered
K_{frak} as tempering instrument	[X] no comma in the residue
Harmonic series: primes as generators	[K] physical evidence
Weighting by the exponents (2, 1, 4)	[X] Z-1 decided
ξ shifts the critical line	[X] functionally fixed
Relaxation circumvents the limit	[X] potential must carry it
FFGFT solves the Riemann hypothesis	[X]

The most honest sentence. Four statements hold as **[K]**, and they fall into two pairs. Two are *identities* of the framework: the factorisation of the torus spectral zeta and the symmetry position of the Casimir interval. Two are *measurements* on the prime structure: the bicoherence-measured cross-term situation and its physical evidence in the harmonic series. None of the four says anything about the *location* of the zeros; they say how the primes compose and where the zeta stands in the spectrum of the framework.

The constructive link, by contrast, reduces to $\hbar = 1$ and carries nothing specific. What remains there is a structural statement: the zero density requires a dilation generator, and that lives in the unrolled domain.

What a viable approach would have to achieve: a non-trivial boundary condition on the unrolled scale direction that (a) carries infinitely many modes – hence is not a finite cutoff window – and (b) produces GUE statistics rather than equidistance. Both requirements point in the same direction: the structure must be arithmetic, not geometric.

.8 The harmonic series as physical evidence

The control case of the musical spiral has so far been treated purely arithmetically. It has a physical counterpart that confirms the measured structure independently: the natural harmonic series of a valveless wind instrument. An alphorn has neither valves nor finger holes; it plays the series $n f_0$ unaltered, as a direct image of the natural numbers (z11_alphorn.py).

Primes as generators

[K] Among the natural tones, exactly the prime orders introduce tonal qualities that cannot be composed from lower ones. The composite orders are combinations of what is already present: $4 = 2 \cdot 2$ is the octave of the octave, $6 = 2 \cdot 3$ the octave of the fifth, $9 = 3 \cdot 3$ the fifth of the fifth.

n	octave-reduced	Cents	Role
2	1.00000	0.0	prime – new axis
3	1.50000	702.0	prime – new axis
4	1.00000	0.0	$2 \cdot 2$ – combination
5	1.25000	386.3	prime – new axis
6	1.50000	702.0	$2 \cdot 3$ – combination
7	1.75000	968.8	prime – new axis
9	1.12500	203.9	$3 \cdot 3$ – combination
11	1.37500	551.3	prime – new axis

This is the Euler product structure, made audible: each prime stands on its own, the composites are products. **The bicoherence test measures the same structure** (Section .6) in the Weil spectrum – there coupling only at prime powers, here new tonal quality only at primes. The acoustic and the number-theoretic finding are independent of one another and agree.

Amplitude decay as an acoustic Tenney analogue

[K] High natural tones are harder to sound and carry less amplitude – with rising order the moved air mass decreases. Formally this corresponds to the Tenney height of Section .6, which weights a ratio by $1/(nd)$: both scales fall like $1/n$. The physical reason (decreasing air mass) and the number-theoretic one (increasing harmonic distance) lead to the same form. The high primes colour the timbre but do not carry the foundation – exactly as the heavy points of the weighted Tonnetz lie on the axes while the distant combinations carry no weight.

The same structure in the mass sector

The harmonic series indexes its modes by integers n_i ; the mass formula of the corpus does the same. Doc. 006 sets

$$\xi_i = \xi_0 \cdot f(n_i, l_i, j_i), \quad E_{\text{char},i} = \frac{1}{\xi_i},$$

with quantum numbers n_i, l_i, j_i from the spatial wave equation and rational factors. There the particles are explicitly conceived as *discrete resonance modes* – the same language as for the natural tones, and the same indexing by integers.

[K] The factors can therefore be read like Tonnetz coordinates. For the leptons, by way of example:

Particle	r_i	Factorisation	Limit
Electron	4/3	$2^2/3$	3
Muon	16/5	$2^4/5$	5
Tau	25/9	$5^2/3^2$	5

The lepton generation thus lies in the same 5-limit lattice as ξ itself ($\xi = 2^2/(2^4 \cdot 3 \cdot 5^4)$). In the quark sector higher primes occur occasionally; whether these are genuine axes or artefacts of the factor fitting is a question of the mass sector and is to be treated there.

[B] Reference. The mass formula, the assignment of quantum numbers and its numerical verification are documented in full in Doc. 006 [13]; the scripts there carry out the reconstruction for all particles. The present document refers to it only to record the structural parallel: integer indexing, rational factors and an octave respectively generation hierarchy

are the same form in acoustics and in the mass sector. No statement about the Riemann zeta follows from this – here the primes appear as *building blocks* of ratios, not as lengths of a spectrum.

Winding number, not frequency ratio

[K] For orbits on the torus what counts is the *winding number* – the fraction of the cycle, hence the logarithm to the octave base – not the frequency ratio itself. The two must be distinguished:

Fifth	Ratio	Kind	Winding number	Kind
pure 3/2	1.5000000	rational	0.5849625	irrational
tempered	1.4983071	irrational	0.5833333	rational = 7/12

The check decides: twelve pure fifths, octave-reduced, end at 1.013643 – the residue is the Pythagorean comma (23.46 cents). Twelve tempered fifths end at exactly 1.000000000000.

The assignment

Pure natural tones have irrational winding and a dense, open orbit. Tempered tones have rational winding and a closed orbit. The closed circle of fifths is an artefact of tempering; in pure intonation it is a spiral. This is the same dichotomy as in Section .2: rational winding $\Rightarrow$ closure.

Cent bookkeeping

[B] The theoretical deviation of a natural tone from the tempered scale and the instrument-induced shift are two different quantities; only their sum is the net deviation. The separation is necessary because the two are easily conflated:

n	Ratio	theor. dev.	Instrument	Net
4	1.00000	0.0	0.0	0.0
7	1.75000	−31.2	−12.5	−43.7
11	1.37500	−48.7	+20.0	−28.7

The 7th natural tone already deviates by −31.2 cents from the tempered seventh; the bell geometry pulls it further down. The 11th lies −48.7 cents below the tritone in theory but is raised again by the cone and the player's embouchure. The instrument values are order-of-magnitude figures from instrument acoustics, given here only to illustrate the bookkeeping.

.9 Averaging, tempering, resolution: where does the instrument sit?

Euler's spiral closes only by tempering (Section .2), and the 5-limit contains the higher limits only by approximation, resolution or tempering (Section .6). In both cases closure arises not from the structure but from something added from outside. It is worth determining what that addition is – for in three domains it sits in three different places.

Three domains, three locations

For hearing, the averaging sits in the *receiver*: critical bands, finite frequency resolution, fusion of closely spaced tones. The measuring instrument becomes part of the result. Tempering here is not an arbitrary stipulation but follows from a property of the listener.

In pure mathematics there is no receiver. The ratios exist regardless of whether anyone distinguishes them. If something is too close that does not close of itself, tempering must be inserted as a *stipulation* – a deliberate intervention, documented in the comma that thereby disappears.

In nature it is open whether there is a receiver. The question therefore arises whether the fractal correction plays that role.

The testable form

[B] A tempering always leaves a residue, and that residue is a *comma*: a ratio of smooth numbers, that is, of products of small primes. The corpus knows one residue – the ≈ 7 eV of P-315-2, 45,000 times the measurement floor. The question is thus testable: does this residue have comma structure?

[K] It does not. The best rational approximations to $1 + 1.36 \cdot 10^{-5}$ are $73530/73529$, $882365/882353$ and $1250017/1250000$; none is smooth. The control probe shows the test's discriminating power: all six commas examined – Pythagorean, syntonic, schisma, breedsma, ragisma, landscape – are recognised as 5- or 7-limit smooth.

On test conduct: the smoothness test must be coupled to a condition on approximation quality. Otherwise the trivial approximation $1/1$ counts as smooth although it misses the target value by 100 %.

Where the fractal correction stands

K_{frak} is not the tempering instrument

If K_{frak} were a tempering, it would have to leave a comma-like residue. The residue present does not have that structure. **[X]** This sharpens P-315-2: the ≈ 7 eV are more likely a pole-mass effect or a genuine correction term than the by-product of a rationalisation.

[B] A further clarification is easily overlooked: in FFGFT the stipulation sits not at K_{frak} but one level below – at $\xi = 4/30000$ itself. That is the rational intervention; $K_{\text{frak}} = 74/75$ is its *consequence*, not the act. In this respect the fractal correction is not a tempering but the bookkeeping of an already closed cycle.

[B] Nevertheless it sits structurally where an instrument would sit: Doc. 314 Ch. D2 carries K_{frak} as a *property of the traversal*, entering via the bridge constants rather than the local operator. That is literally “the measuring instrument becomes part of the result”: K_{frak} lies not in the field but in the traversing. It resembles a measurement effect in *location*, not in *character* – it is exact, not averaging.

The actual candidate: cutting off rather than averaging

[B] If nature has an instrument, it is more likely the double cutoff. It generates a resolution limit, and precisely there the procedure becomes part of the result – without anything being tempered. Section .6 shows it in a concrete case: at a peak width of 0.0021 the 5-limit approximation to $\ln 7$ is no longer distinguishable from the exact value, although the ragisma between them remains *present*.

The difference in one sentence

A tempering *alters* the values and leaves a comma. A resolution limit alters nothing and leaves nothing – it merely makes differences invisible. Both make distinctions disappear; only the first produces residues.

Hence the tripartition. Hearing averages *actively* and produces commas. Mathematics has no receiver; there tempering must be done by stipulation. Nature, if the framework is right, cuts off *passively*: distinctions vanish without rationalisation – and the fractal correction is not involved in this; the scale limit is.

[B] Status. The comma test is an exclusion and rules out one particular reading of K_{frak} . It does not show that no averaging mechanism acts in nature at all – only that the fractal correction is not one. Whether the double cutoff is more than an analogy as an instrument remains open; it depends in turn on quantities the corpus carries as open (the prefactor 4 in L^* , R73).

.10 Sources and methodological placement

The procedures used in this document are standard, and so are their pitfalls. This section places the usable literature and states where the reach of the procedures employed is bounded.

Load-bearing sources

Berry & Keating (1999) [1]. The basis of Section .4. Comparison of the counting functions suggests that the t_n are eigenvalues of an unknown Hermitian operator obtained by quantising a classical system with Hamiltonian $H_{cl} = xp$. The work also states the requirement on such a system: the “Riemann dynamics” would have to be chaotic and possess periodic orbits *whose periods are multiples of logarithms of the primes*. That is precisely the length spectrum of Section .6 – and the condition on which the Tonnetz fails, since it yields sums rather than multiples.

Sierra (2019) [2] and **Sierra & Townsend (2008)** [3]. These show the state of the Hilbert-Pólya programme: spectral realisations via xp with boundary interaction, via Landau levels, and most recently via a massless Dirac fermion in Rindler spacetime with delta potentials on the square-free integers. Notable for the present question: all these constructions encode the arithmetic *explicitly* (square-free numbers, modular forms, boundary functions from the Riemann-Siegel formula). None derives it from a geometry. This supports the finding of Section .3 from the other side: where the zeros are realised spectrally, it happens by building in number theory, not by choosing a space.

Pápics (2012) [4]. The work methodologically closest to Section .6. It examines whether detected frequencies stand in harmonic relation to one another – the same problem in asteroseismology. Central result: combination frequencies occur in random data with high probability, and as data quality rises the distinction between genuine and spurious combinations becomes *harder*, not easier. The usable discriminator is not the existence of individual combinations but the *number* of lowest-order combinations compared against simulations with matching properties. Exactly this approach – comparison against a simulated null distribution rather than evaluation of individual hits – underlies the permutation tests of this document.

Pólóskei et al. (2018) [5]. Treats the procedure actually responsible for the question of harmonic coupling. Bicoherence

$$b^2(f_1, f_2) = \frac{|B(f_1, f_2)|^2}{A\{|X(f_1)X(f_2)|^2\} A\{|X(f_1+f_2)|^2\}}$$

measures not amplitudes but *phase coupling*: whether $f_3 = f_1 + f_2$ occurs with a fixed phase relation $\varphi_3 = \varphi_1 + \varphi_2 + \text{const.}$ Decisive for the present use is the warning about false positives: for non-stationary or short signals, classical bicoherence yields high values *without* any coupling. The proposed remedy is a Monte Carlo procedure using the measured Fourier amplitudes with *randomised phases*, from which the null distribution $\rho(b)$ and a critical value $b_c(\alpha)$ are obtained. The trade-off is also stated: the number of averages can be increased only at the cost of frequency resolution ($\Delta f = f_s N / M$).

Further usable procedures

The following methods fall below the Fourier resolution limit by assuming a model (a sum of exponentials) instead of transforming blindly. They have not been employed here but would be the next stage:

- **Subspace methods:** MUSIC (Schmidt), ESPRIT (Roy/Kailath), Min-Norm. They decompose the covariance matrix into signal and noise subspaces.
- **Matrix pencil [7]** for damped and undamped sinusoids.
- **Iterative methods:** RELAX, NOMP; and Volterra-series approaches for the self-consistent removal of combination frequencies (Lares-Martiz et al., MNRAS 2020).
- **Multitaper** (Thomson) with an F-test for line components; **multichannel singular spectrum analysis** for extracting dynamical modes without a Fourier basis.

On the magnitude of the gain: in a published comparison of three frequencies at 440/445/450 Hz over 100 ms, ESPRIT attains a frequency RMSE of 0.087 Hz against 2.88 Hz for spectral MUSIC – the difference between model classes thus exceeds the difference from Fourier analysis.

Reach of the procedures employed

[B] The scripts z6 and z7 use the bare Fourier sum $F(u) = |\sum_n e^{iy_n u}|/N$ – the simplest tool available. Three limitations follow:

1. *No coupling measurement.* $F(u)$ measures whether structure sits at length u at all, not whether two lengths are *coupled*. The structural claim that the Euler product has no cross terms is argued in this document from the form of the product and measured in Section .6; the remaining tests of this document manage with the simpler procedure.
2. *Statistics.* Sections .6 and .6 compute with 100 zeros from the literature; for bispectra that is too few. Section .6 therefore computes 2469 zeros itself.
3. *Not a time signal.* The sequence of zeros is not a measured signal; stationarity is undefined, and the standard tests for non-stationarity do not apply directly.

These limitations do not touch the **[X]** findings of this document: the Weyl obstruction (Section .3) and the rigidity of the Weil weights (Section .6, Test C) are structural arguments and independent of the choice of spectral method. They concern only the numerical tests, whose force is thereby bounded from below – which does not weaken the negative overall finding but merely qualifies its numerical component.

.11 Closing reflection: why one cannot compute backwards

At the end stands an image that draws the individual findings together, and an interpretation that goes beyond them. The two must be kept apart.

What the calculations show

The primes are axes – in the Tonnetz as in the Euler product. The same numbers, the same lattice structure. For simple ratios the inference back to the underlying harmony is unambiguous: $\ln 2$, $\ln 3$, $\ln 5$ appear as clear lines in the spectrum of the zeros, each in its place, each assignable.

With composition this unambiguity is lost, for two reasons, both visible in the tests.

The first is *ambiguity*. A simple ratio has few possible decompositions. A composite one has many – and the further the lattice is spanned, the denser the possibilities lie, until in the end every number fits somehow. It is not the rarity of a combination that creates the difficulty but its multiplicity of readings: where everything is reachable, reaching says nothing.

The second reason weighs heavier. The Euler product treats each prime *on its own*, without cross terms; its spectrum knows only $k \ln p$, multiples of individual prime logarithms. As soon as a ratio mixes several primes it is no longer a single harmony but a superposition – and superpositions do not appear in the spectrum as lines of their own. $\xi = 2^2/(2^4 \cdot 3 \cdot 5^4)$ is such a superposition.

An image from practice captures it exactly: from the sound of a single reed one can infer the fundamental. In a full chord across several registers one hears the mixture – but the fundamental of the mixture does not exist. There are only the fundamentals of the individual reeds. ξ is a chord, not a note.

The direction that works – and the one that does not

This allows a precise statement of what is actually closed here. It is not the connection between harmony and structure; it is its *reversibility*.

Forwards it carries: from simple ratios the composite can be built, step by step, each step traceable. This is how intervals arise from fifths and thirds, and how ξ arises from 2, 3 and 5.

Backwards it does not carry. Composition is not invertible because it destroys information: many paths lead to the same place, and the path taken can no longer be read off the result. This is not a weakness of method that better techniques could remedy, but a property of the matter itself. The tests of this document confirmed it at three different points: at the weighting strength, at the limit extension, and at the rigidity of the Weil weights.

Interpretation

Interpretation, not a finding

It is natural to suppose that complex formations in nature rest on such harmonies – that the composite is built from simple ratios, as a chord from notes. This conjecture is old and has been fruitful in many respects; it also stands behind the construction of ξ .

What this document shows is something else and more modest: even if that is so, the path cannot be walked backwards. Reconstructing the underlying harmonies from

the finished formation is not difficult but impossible – composition has erased the assignment.

The first sentence is an interpretation and remains one; nothing in this document supports it and nothing refutes it. The second is computed.

Between the two lies the actual yield: whoever suspects a harmonic order behind complex structures must establish it *forwards* – by construction and prediction. The way back, inferring the order from the finished structure, is not open. This holds for the Riemann hypothesis, and it holds more generally.

.12 The Fourier transform as a projection – and why that settles it

This document has applied the Fourier transform to 100 and 2469 zeros respectively (Sections .6 and .6) and made prime lengths visible in the process. It is worth understanding that operation precisely – for it reveals why neither classical FFT, nor quantum Fourier transform, nor a photonic system can solve the Riemann hypothesis.

What a Fourier transform does and does not do

[K] A Fourier transform – classical, optical, or as a quantum Fourier transform (QFT) – is a *projection*: it translates one representation into another without generating new information. What is not contained in the input data does not appear in the output.

This is visible concretely in the measurements of this document. The Fourier transform of the zeros shows peaks at $\ln 2$, $\ln 3$, $\ln 5$, $\ln 7$, $\ln 11$, $\ln 13$ – because this information is *already contained in the zeros*, via the Euler product. It shows no peak at $\ln(1/\xi) = 2 \ln 2 + \ln 3 + 4 \ln 5$ – because that information is not there. The transform *invents* no structure; it *projects* the existing structure onto a different coordinate system.

Consequence: A Fourier analysis of the zeros cannot solve the Riemann hypothesis, because it does not generate the arithmetic information – it can only find it if it is already there. And the decisive information (exactly where the zeros lie) must be obtained by computation, not by projection.

The structure of Shor's algorithm

[K] Shor's algorithm decomposes factorisation into two steps. The first – evaluating $f(x) = a^x \bmod N$ – is classical overhead that cannot be parallelised; it grows with bit size. The second – the QFT for period finding – is the actual quantum operation.

But the QFT too *generates* no new information: it finds the period r already present in the prepared states. *Obtaining* those states is the limiting step. The same holds for the zeros: the Fourier analysis finds what the Riemann-Siegel formula has already computed. The QFT does not solve the RH because it does not compute the zeros – it takes them as given.

The topological obstruction

[K] Section .3 shows the decisive obstruction: the density of zeros grows like $T \ln T$, while every compact manifold (and every finite qubit register) has eigenvalue density $N(\lambda) \sim C \lambda^{d/2}$. That is not a power – and hence structurally incompatible. A quantum computer with n qubits spans a compact, finite state space; it cannot carry the $T \ln T$ density as an eigenvalue spectrum. This is not a technical but a structural obstacle.

The connection to Shor: Shor's algorithm factorises concrete numbers efficiently on quantum hardware – a real achievement for a specific problem. But a general exponential advantage of quantum computers over classical methods is not established, and for the RH the Weyl obstruction applies without qualification.

Projection and counting

The Fourier transform operates in the additive space of logarithms; the Riemann hypothesis is a statement about the multiplicative structure of the Euler product. A projection translates the representation – it does not change the question.

The core of the separation

The Fourier transform is a projection of multiplicative structure onto an additive basis. It can *find* the prime structure when it is present in the input data – as the bicoherence test demonstrates. It cannot *generate* it. The Riemann hypothesis is a statement about the consistency of factorisation – and that question cannot be answered by any projection, whether classical, optical, or as a quantum Fourier transform.

Note on the Shor documents in the corpus. Older documents (Docs. 075, 076, 147, 176) draw a connection between FFGFT and Shor's algorithm. That connection is associative, not substantive: factorising a particular number is structurally distinct from a statement about all primes. Those documents were revised in August 2026 accordingly (R75, Doc. 190).

.13 Relaxation as a third mechanism

The preceding sections treat procedures that *transform*: Fourier transform, quantum Fourier transform, optical interference. All are projections and generate no information (Section .12). There is a third mechanism that does not transform but *settles*: physical relaxation into an energy minimum.

The mechanism

[Q] An Ising system minimises

$$H_{\text{Ising}} = -\frac{1}{2} \sum_{ij} J_{ij} \sigma_i \sigma_j - \sum_i h_i \sigma_i, \quad (1)$$

and every combinatorial optimisation problem maps polynomially onto this form. Whoever finds the ground state has solved the problem. Analog Ising machines – for instance coherent optical setups of degenerate parametric oscillators – realise this as physical relaxation rather than algorithmic iteration. The corpus treats this approach in Doc. 161 (T0 Ising machine) and Doc. 162 (comparison of optimisation frameworks).

The method is established. Programmable setups with 100 spins and all-to-all coupling, as well as networks with 2000 nodes, are published (McMahon et al., *Science* 354, 2016; Inagaki et al., *Science* 354, 2016).

Noise as a tool

[Q] The central difficulty of analog Ising machines is getting trapped in local minima. Noise resolves this: it forces spin flips even where they are energetically unfavourable, opening a

wider region of phase space. This is not conjecture but measured practice with a quantitative law.

Decisive is an *optimal* noise strength: too little leaves the system trapped, too much drives it into a Boltzmann sampling regime in which the ground state is likewise missed. The optimal strength follows a scaling law

$$\gamma \sim \beta\sqrt{C}, \quad (2)$$

with coupling strength β and mean number C of connections per spin; the fit reaches $R^2 = 0.97$ for small problems (Mys, Verschaffelt & Van der Sande, arXiv:2508.19107, 2025). This allows the quality to be predicted without a parameter scan – the times to solution obtained lie at about twice the optimum.

What relaxation achieves and what it does not

[K] Relaxation is *not* a projection. It does not transform between representations but settles into a state. The exclusion of Section .12 therefore does not apply directly; the mechanism must be examined in its own right.

[X] It fails at the same point, but for a different reason. The potential J_{ij} must *already* carry the sought structure: the optimisation problem is encoded into the coupling matrix, and relaxation finds what is written there. An arithmetic structure not contained in J_{ij} arises no more from settling than from transforming.

There is also a practical limit that the sources themselves state: not all problems reach high success rates. In the study cited, five of the forty Gset instances yield, for *none* of the models examined, a value within 99.5 % of the best known cut. Analog Ising machines are capable heuristics, not exact solvers – the ground-state problem remains NP-hard.

Three mechanisms, one limit

Transformation (Fourier, QFT, optics) projects and generates no information. Relaxation settles and finds what is in the potential. Both presuppose that the sought structure is already present – in the input data or in the coupling matrix respectively. The Riemann hypothesis is a statement about that structure itself.

The question, properly posed

This allows the initial question to be put more sharply. “Can a quantum computer solve the Riemann hypothesis?” confuses the tool with the object – and forces every technology to be examined separately.

The tenable formulation reads: *Is the Riemann hypothesis a question that can be answered by transformation or relaxation?* The answer is no, and it holds independently of technology: both mechanisms presuppose the arithmetic structure rather than generating it. The limit lies not in the tool but in the object.

.14 Settled items

The questions raised in the course of this document are all decided. They are recorded here so that they are not taken up again.

- **Z-1 – weighting by the exponents** (2, 1, 4): decided in the negative. Neither a weighting strength nor a limit extension produces selection; above all, the weights of the explicit formula are not free parameters (Section .6).

- **Z-2 – meaning of the lattice factor** $(1 - 4^{1-s})$: clarified. The factor is the Dirichlet signature of Jacobi's four-square theorem, $r_4(n) = 8 \sum_{d|n, 4 \nmid d} d$, and encodes the divisibility condition "4 does not divide d ". The associated ladder of zeros $s = 1 + 2\pi i k / \ln 4$ with spacing $\pi / \ln 2 = 4.5324$ is a property of the sum of four squares; the nearest quantity of the framework (the prefactor 4 from L^* , R73) is 12 % away. The 4 is indeed the number of squares and hence the dimension, but it acts arithmetically here, not geometrically (z12_gitterfaktor_waermekern.py).
- **Z-3 – heat kernel and theta-series fine structure**: void. The difference between D4 and $\mathbb{Z}^4$ rests on a parity condition – D4 occupies only even norms, and at those the two lattices agree exactly. No test surface follows: at small t the Weyl term dominates both series equally, at large t both are exponentially small, and the informative location is the Casimir point $s = -1/2$, whose evaluation is given in Doc. 314.
- **Z-4 – bicoherence test**: carried out and confirming in six of six controls (Section .6).
Not recommended: further search for ξ periods in the zeros. Section .5 excludes this structurally via the GUE argument, not merely numerically.

.15 Verification

All numbers are reproducible by twelve scripts (standard library only, target values as assertions, seed 20780458):

- z1_weyl_obstruktion.py – Weyl's law against Riemann-von Mangoldt, local exponent, power fit.
- z2_zellbedingung.py – Berry-Keating count, equivalence to the Riemann form, cross-check against Weyl.
- z3_randbedingung_test.py – ξ ladder test with artefact control, null distribution, out-of-sample, GUE control.
- z4_epstein_casimir.py – factorisation against the direct sum, Casimir symmetry point.
- z5_zellbedingung_kritik.py – examination of the cell argument in three stages.
- z6_tonnetz_weil.py – Tonnetz coordinates of ξ , Fourier detection of the Weil length spectrum, resolution limit.
- z12_gitterfaktor_waermekern.py – Jacobi's four-square theorem as the origin of the lattice factor; shell occupation of D4 against $\mathbb{Z}^4$.
- z11_alphorn.py – harmonic series, winding number versus frequency ratio, cent book-keeping.
- z10_kommatest.py – does the residue of P-315-2 have comma structure? Smoothness test with control probe.
- z9_limit_einschluss.py – five readings of limit inclusion; ragisma and Archytas' comma as approximation errors.
- z8_bikohaerenz.py – zeros via Riemann-Siegel, bicoherence with phase randomisation, positive and negative controls.
- z7_gewichtung_unmoeglich.py – weighting strength α , limit extension (5-, 7-, 11-limit), permutation tests, rigidity of the Weil weights.

Inputs: $\xi = 4/30000$ exact; $\ell_P = 1.616255 \cdot 10^{-35}$ m, $\bar{\lambda}_e = 3.8615926796 \cdot 10^{-13}$ m (CODATA 2022); zero values (first 50) from the literature.

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